

Coding & Solution

Date	02 October 2026
Project ID	SWTID-2026-8542
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of the implemented code and the overall solution. ComicCraft demonstrates an end-to-end workflow for generating personalized comic stories, illustrations, previews and downloadable PDF comics using FastAPI and AI models.

Instructions:

1. Ensure the code is well-structured, commented, and follows best practices for the chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to the code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	Not provided in the supplied project document
Programming Language(s)	Python, HTML, CSS
Framework(s) Used	FastAPI, Jinja2; Google Gemini API; Hugging Face Diffusers / Stable Diffusion; FPDF
Key Features Implemented	• User inputs: story prompt, character name, setting, tone and art style. • Gemini Flash generates a structured 5-panel outline. • Gemini Pro generates narration and character dialogue. • Stable Diffusion generates comic-style illustrations. • Comic preview displays panels with title, image, description and narration. • FPDF exports the complete comic as a downloadable PDF. • FastAPI routes support form and JSON-based comic generation.
Pending / Incomplete Features	User account management is not included in the current version. The supplied document identifies future enhancements such as user profiles, comic libraries and multi-page story arcs.
Setup / Run Instructions	1. Install Python and pip. 2. Create and activate a virtual environment. 3. Install dependencies using <code>pip install -r requirements.txt</code> (or the documented package list). 4. Configure <code>GEMINI_API_KEY</code> and <code>HF_API_KEY</code> environment variables. 5. Start the server with <code>uvicorn app.main:app --reload</code>. 6. Open <code>http://127.0.0.1:8000</code>; API docs are available at <code>/docs</code>.

Source basis: Prepared from the supplied ComicCraft project document and formatted to match the two-page SmartBridge / SkillWall reference layout. No repository URL was supplied, so none was invented.

Code Quality Checklist

S.N O	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Not stated
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Not verified
6	Code is committed to a version control repository	Not stated

Additional Notes / Comments:

Project: ComicCraft - AI Comic Story Creator using Gemini Models

The supplied project document describes a modular FastAPI application with separate functions/files for outline generation, story generation, image generation, layout building and PDF export. It also documents FastAPI error handling using HTTPException and try-except blocks. Items marked "Not stated" or "Not verified" should be confirmed by the project team before final submission.